

Devam Mondal

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EDUCATION

Georgia Institute of Technology

May 2028

BS in Computer Science | Concentrations: Intelligence + Modeling and Simulation | GPA: 3.94

Atlanta, GA

- **Coursework:** Data Structures & Algorithms, Objects & Design, Artificial Intelligence, Machine Learning, Computer Vision, Computer Organization, Computer Systems & Networks

TECHNICAL SKILLS

Languages & Developer Tools: C++, Python, Java, C, JavaScript, Git, Docker, Linux, Vercel, Terraform

Frameworks & Libraries: Flask, FastAPI, Django, React, Node.js, Express.js, OpenCV, LangChain, HuggingFace

Databases & Cloud Platforms: MongoDB, SQL, Firebase, AWS EC2 + SageMaker + S3 + SQS, Google Cloud

Concepts: Agile, REST APIs, Databases, LLMs + RAG, Distributed Systems, Information Retrieval, Concurrency

EXPERIENCE

Amazon Web Services (AWS)

Sept 2026 – Present

Software Development Engineer Intern

Seattle, WA

- Prototyping an MCP server enabling low-latency, natural-language queries across **1M+** DNS and DHCP resolution log records, streamlining on-call network troubleshooting.
- Shipped AWS Lambdas to convert JSON logs to Parquet and register schemas, improving Athena query speed by **10x**.

Millennium

Jun 2026 – Aug 2026

Software Engineer Intern

Miami, FL

- Engineered a C++ gRPC service to serialize and stream **50+** Apache Arrow Level 1 order book snapshots/sec over TCP into downstream ingestion pipelines, accelerating strategy backtesting on sparse market data.
- Optimized Apache Arrow table-processing throughput by **5x** using multithreaded workload partitioning and semaphore-based synchronization.
- Reduced string lookup latency by **20%** by replacing unordered maps with thread-local vectors for better cache locality.

Cisco

May 2025 – Apr 2026

Software Engineer Intern

Atlanta, GA

- Prototyped and deployed **4** Python API endpoints with Terraform and Jenkins to explore Gmail message remediation for the Secure Email Threat Defense team, expanding product reach.
- Implemented a bulk remediation and reclassification pipeline using two AWS SQS queues and AWS Lambda workers, deploying to beta production and empowering administrators to process **5,000+** emails concurrently.
- Engineered a time-throttle mechanism for bulk remediation requests using an AWS DynamoDB audit table, increasing backend reliability and reducing system load spikes by **15%**.

InPharmD (YC W21)

Feb 2025 – Apr 2025

Artificial Intelligence Engineer Intern

Atlanta, GA

- Designed a novel deep-research agentic workflow with TF-IDF document search using OpenRouter, MongoDB, and AWS EC2, generating **20+** page drug reports for pharmacists through LLMs and cutting diagnosis times by **80%**.

PROJECTS

C++ Transformer w/ Inference Endpoint | C++, Eigen, Crow

- Implemented a Transformer language model in C++ and corresponding inference endpoint, including single-head attention, cross-entropy loss, and layer normalization, using wrapped Eigen matrix operations.
- Built a custom word-level tokenizer, constructing vocabularies of **5,000+** tokens from **10,000** lines of raw text.

Toffee | React, FastAPI, Firebase, Firestore, OpenRouter, Vercel

- Developed a full-stack AI study tool with React and FastAPI that converts documents into quizzes and flashcards.
- Designed five API endpoints to convert PDFs, plaintext, and YouTube transcripts into JSON-formatted quiz questions using Google Gemini Flash and Regex, deployed on Vercel.

Contacts | Python, FastAPI, OpenAlex, MongoDB, BM25, Vercel

- Built a FastAPI research discovery tool that generates topic knowledge graphs from academic search terms by traversing works, references, and related publications across journals through the OpenAlex API.
- Engineered a multithreaded graph expansion pipeline with concurrent search requests, enabling discovery across **400+** papers per query while ranking candidates with BM25 relevance scoring.